

No Simple Answer to Data Complexity: An Examination of Instance-Level Complexity Metrics for Classification Tasks

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Abstract

Natural Language Processing research has become increasingly concerned with understanding data quality and complexity at the instance level. Instance-level complexity scores can be used for tasks such as filtering out noisy observations and subsampling informative examples. However, there exists a diverse taxonomy of complexity metrics that can be used for a classification task, making metric selection itself difficult. We empirically examine the relationship between these metrics and find that simply storing training loss provides similar complexity rankings as other more computationally intensive techniques. Metric similarity allows us to subsample data with higher aggregate complexity along several metrics using a single *a priori* available meta-feature. Further, this choice of complexity metric does not impact demographic fairness in downstream predictions. We encourage researchers to carefully consider metric availability and similarity, as using the wrong metric or sampling strategy may hurt performance.

1 Introduction

Understanding data complexity at the instance-level has become increasingly important in Natural Language Processing (NLP) and machine learning (ML). Recent work has shown that model performance can be improved through dataset curation, curriculum learning, and in-context learning techniques (Smith et al., 2014; Toneva et al., 2019; Shen and Sanghavi, 2019; Lu et al., 2023). To perform these techniques, researchers need some measurement of data complexity taken at the instance-level to logically filter or order data observations.

NLP tasks are particularly well-suited for studying this area due to the complexity of language and its impact on classification results (Ethayarajh et al., 2022; Gururangan et al., 2018; Hahn et al., 2021). Indeed, these complexity-based techniques are increasingly being used for data filtering and

re-weighting in NLP tasks including text classification, text generation, and question answering (Rodriguez et al., 2021; Lalor et al., 2019; Soviany et al., 2022).

Due to its close connection to misclassification rate, instance complexity also has important implications for how we think about bias and fairness (Lorena et al., 2024). Prediction differences across subgroups have the potential to increase harm for underprivileged groups in certain systems (e.g., hiring decisions and facial recognition, Li et al., 2023; Lorena et al., 2024). Bias can exist in data, algorithms, and prediction outputs; addressing fairness across the ML pipeline can mitigate harm (Pessach and Shmueli, 2022; Lalor et al., 2024).

Any technique that alters the training data risks amplifying algorithmic bias (Zhao et al., 2018; Blodgett et al., 2020). If a selected complexity metric disproportionately removes data from certain protected subgroups, this under-representation bias will be captured by the model. Within the current Large Language Model (LLM) paradigm, the addition of complexity-based techniques risks exacerbating existing data biases from pretraining data content (Li et al., 2020; Abid et al., 2021), human annotation (Kirk et al., 2023; Gururangan et al., 2018), and user feedback (Qiu et al., 2022).

Prior work proposed a taxonomy of data complexity for classification tasks (Lorena et al., 2024). However, it is not always clear to researchers which metric they should use to leverage instance-level complexity in their experimental design (Paiva et al., 2021; Martínez-Plumed et al., 2019). Metrics are available at different points of the learning process (e.g., data preprocessing, embedding, training, etc.) and require varying amounts of computation.

To address this gap, we seek to understand how different instance-level complexity metrics relate to one another and if there is any overlap in their measurement. We train a pool of text classifiers, compute a diverse set of complexity metrics, and

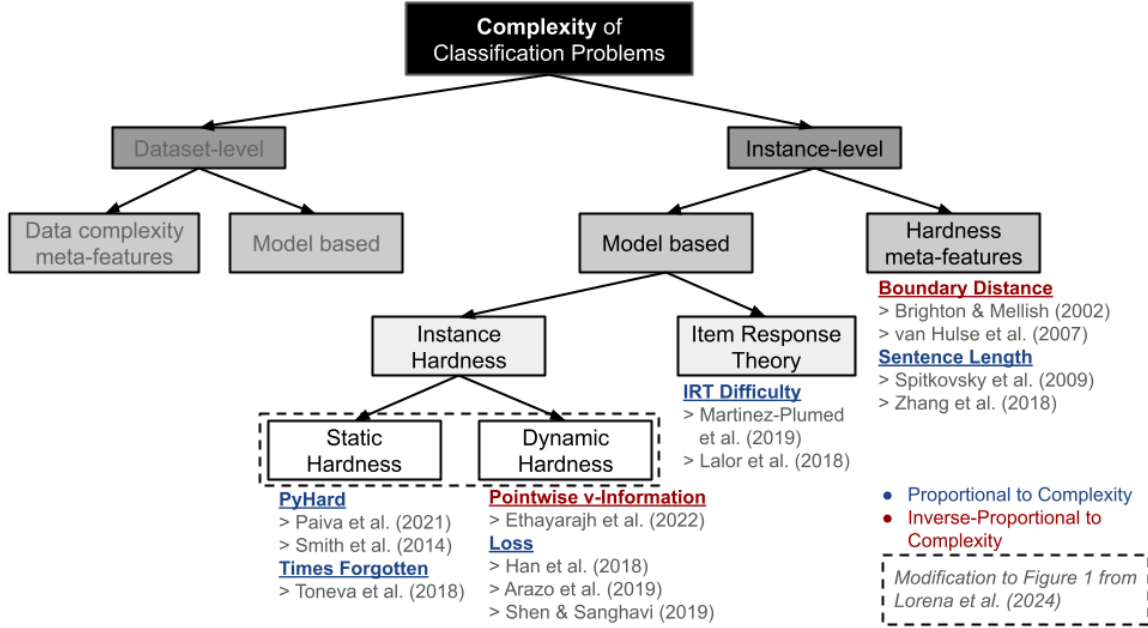


Figure 1: Classification complexity taxonomy tree introduced in Lorena et al. (2024) with our modifications. Note that in this paper we focus on the instance-level, i.e., the right-hand-side of the tree.

examine how these complexity metrics correlate across different classification tasks. We also apply a naive *a priori* subsampling strategy using a complexity metric available before model training to examine whether including more complex observations might improve performance.

Our contributions in this work are: (i) an extended taxonomy of complexity, as well as a complexity analysis of various techniques; (ii) an empirical examination of the relationships between different complexity metrics across models and dependent variables; (iii) an examination of the combination of complexity and fairness when considering data complexity split by demographic groups.¹

The rest of the paper is organized as follows. Section 2 reviews work related to instance-level complexity and the taxonomy we extend. We explain these updates in Section 3 along with the representative metrics used in our experiments in Section 4. We discuss findings in Section 5 before offering concluding remarks in Section 6 and limitations in Section 7.

2 Related Work

Research has long sought to quantify the complexity of text sequences using linguistic measures such as syntactic dependence and semantic entropy (Gib-

son, 1998; Hale, 2001). Instance-level complexity has remained an NLP problem with the rise of computational techniques concerned with which text sequences are harder to learn (Zhao et al., 2022; Zhang et al., 2022; Hahn et al., 2021; Cai et al., 2024). In this section, we begin with a broad view of complexity in ML and refine our focus to NLP applications and an applicable taxonomy of complexity for these tasks.

2.1 Instance Complexity Overview

Complexity literature covers the analysis of both dataset-level and instance-level complexity (Smith et al., 2014; Lorena et al., 2024). Especially in NLP, analysis of dataset-level quality allows for prioritization of certain tasks, benchmarks, and classifiers within the Common Task Framework commonly used for evaluating NLP methods (Ho and Basu, 2002; Donoho, 2017). On the other hand, assigning complexity scores at the instance-level allows researchers to identify problematic instances to be filtered (e.g., outliers, annotation artifacts) and informative instances to prioritize.

A priori filtering of outliers – i.e., before training – can improve performance and generalizability through denoising the data in the data curation paradigm (Smith and Martinez, 2011; Hodge and Austin, 2004). Specifically in NLP tasks such as question answering and machine translation, studies have used human-centered complexity metrics

¹Code and data available at <https://github.com/nd-hal/instance-complexity-metrics>.

to identify annotation artifacts, which are problematic instances resulting from faulty patterns in responses or data labeling from crowd workers (Gururangan et al., 2018; Rondeau and Hazen, 2018). Theory-driven techniques such as Item Response Theory and information theory learn estimates for identifying artifacts and measuring fairness (Lalor et al., 2019; Martínez-Plumed et al., 2019; Ethayarajh et al., 2022).

Beyond fixed measures of *static* complexity, NLP leaderboards increasingly consider *dynamic* complexity, which is calculated across epochs (Swayamdipta et al., 2020; Rodriguez et al., 2021). Since model-based complexity scoring is computationally intensive, re-weighting and dynamic filtering are often accomplished via the current value of the loss function in the curriculum learning paradigm. Complex instances have also been conceptualized as providing lower information to the model or being forgotten in later epochs (Toneva et al., 2019; Ethayarajh et al., 2022). However, most instance-level complexity measures for NLP come from naive linguistic heuristics such as sentence length or number of conjunctions, which are known *a priori* before training (Soviany et al., 2022; Zhang et al., 2018).

2.2 Instance Complexity Taxonomy

Recent work has presented a taxonomy of data complexity for classification tasks, where complexity is defined as “the difficulty level in predictive problems” (Lorena et al., 2024). We take the framework from Figure 1 of Lorena et al. (2024) and focus on the right side of the tree, which considers instance-level metrics; we refer to this as *Instance Complexity* or simply “complexity.” Here complexity refers to intrinsic characteristics (i.e., meta-features) of an instance that increase its likelihood of being misclassified. In this work, “complexity” will serve as a superset for all other related terms such as “hard,” “difficult,” and “challenging.” Further, we extend their taxonomy for our Figure 1 above to (1) make a distinction for the dynamic nature of certain instance hardness metrics and (2) include earlier literature on inter-class boundary distance.

3 Instance Complexity Revisited

Next, we describe our literature search and the corresponding updates to the taxonomy of Lorena et al. (2024). We explain each metric from the updated taxonomy used in our experiment.

3.1 Literature Search

To identify papers, we conducted a review of the literature for work dealing specifically with instance-level complexity. We first identified a set of seed papers, including the Lorena et al. (2024) taxonomy we extend, its foundational work (Smith et al., 2014), and work also dealing with complexity metrics (Ho and Basu, 2002; Soviany et al., 2022). We examined the references of these seed papers and collected 30 metrics from $n = 39$ papers related to Instance-level complexity. We focus our analysis on 7 representative metrics; a full list is in Appendix A for reference.

3.2 Updates to the Taxonomy

Our first change reflects the hierarchical nature of the taxonomy’s “Instance Hardness” category (in the lower middle of Figure 1). Here, certain metrics vary across each epoch and require aggregation to the model-level. These *dynamic* metrics (e.g., Loss) are still “Model-based” as they result from training, but should be considered separately from *static* hardness metrics, which are calculated once at the model-level for the entire training process.

Secondly, classification datasets often generate class variables from transformations of real-valued variables (e.g., Abbasi et al., 2021). This score can be seen as the result of a dimensionality reduction from the real-valued variable to the classification class, but should not be on the “Model-based” side of the taxonomy tree in Figure 1 as it is unrelated to any trained ML model. Instead these measurements fall into “Hardness meta-features” and are intrinsic characteristics of the data. Our intuition that scores closer to a median split should be considered to be more challenging is supported by earlier SVM and neighborhood literature in Section 3.4.

Our updated taxonomy can be found in Figure 1, with proposed modifications in dashed boxes. Below we describe each metric in detail. Metrics colored blue and marked with a $\uparrow$ symbol are positively correlated with complexity, while those colored red and marked with a $\downarrow$ symbol are negatively correlated to complexity. For example, higher IRT $\uparrow$ values indicate higher complexity for instances, while higher PVI $\downarrow$ values show lower complexity.

3.3 Model-based

3.3.1 Static

PyHard (PH) $\uparrow$ Instance Hardness theory contends that the hardness of a given instance is rel-

ative to both the model used to classify it and the complexity of other items in the dataset (Smith et al., 2014). The PyHard algorithm uses instance space analysis to sample only informative meta-features and efficiently generate a single output probability of misclassification (PH) from a pool of seven diverse classifiers (Paiva et al., 2021). Complete computation details can be found in Appendix D, but a higher PH value indicates that an instance has a higher probability of being misclassified:

$$\text{PH}_{\mathcal{L}}(\langle x_i, y_i \rangle) = 1 - \frac{1}{|\mathcal{L}|} \sum_{j=1}^{|\mathcal{L}|} p(y_i | x_i, g_j(t, \alpha))$$

Here $\mathcal{L}$ refers to a pool of diverse learners and $g_j(t, \alpha)$ is the complete set of learning algorithms and their hyperparameters. While the literature recognizes several hardness meta-features for different dimensions of PH (e.g., k-disagreeing neighbors, local set cardinality, etc.), we only consider explicitly this aggregate probability of misclassification (Smith et al., 2014; Arruda et al., 2020; Lorena et al., 2024).

Times Forgotten (TF) $\uparrow$ Forgotten examples are instances which are classified correctly in earlier epochs but misclassified at some later epoch in the training process (Toneva et al., 2019). These instances which are more frequently forgotten are considered to be more complex; we can sum forgetting events across epochs:

$$\text{TF} = \sum_{e=1}^{|E|} \sum_{k=e+1}^{|E|} \mathbf{f}(y_i | x_i)$$

Here $\mathbf{f}(y_i | x_i) = \mathbb{1}(\hat{y}_{i,e} = y_{i,e}) \wedge \mathbb{1}(\hat{y}_{i,e+k} \neq y_{i,e+k})$ indicates a forgetting event at epoch $e + k$.

Item Response Theory (IRT) $\uparrow$ Item Response Theory (IRT) has become increasingly popular, and we increasingly see IRT concepts used in the evaluation of NLP models and datasets (Martínez-Plumed et al., 2016; Rodriguez et al., 2021; Lalor et al., 2016). In a one-parameter (1PL) IRT model, the difficulty b of a given item can be considered as the point on the ability scale θ where the probability of any subject providing a correct answer is $p(y = 1) = 0.5$. Difficulty is estimated from a dataset of graded correct/incorrect responses to questions across subjects to best fit each item’s Item Characteristic Curve:

$$p(y = 1 | \theta, b) = \frac{1}{1 + e^{\theta - b}}$$

Thus, a core principle of IRT is its assumption that the proficiency of a classifier is a function of

the level of hard instances it can solve. Work has been done to scale this IRT parameter estimation to larger numbers of items via Bayesian estimation procedures, so ML applications of IRT can fully take advantage of large text datasets (Natesan et al., 2016; Wu et al., 2020; Lalor and Rodriguez, 2023).

3.3.2 Dynamic

Pointwise v-Information (PVI) $\downarrow$ Pointwise v-Information (PVI) provides an information theoretic perspective on complexity by viewing the difficulty of a given instance as its lack of v-usable information, which considers the accessibility of Shannon mutual information between an encrypted input X and an output Y (Xu et al., 2020; Shannon, 2001). Ethayarajh et al. (2022) extend v-information from dataset complexity to instance-level complexity with PVI:

$$\text{PVI}(x_i \rightarrow y_i) = -\log_2 H_\nu(Y) + \log_2 H_\nu(Y|X)$$

Here $H_\nu(y_i) = \mathbb{E}[-\log g'(y_i | x_i)]$ is obtained from the primary model g' and $H_\nu(y_i) = \mathbb{E}[-\log g(y_i | \emptyset)]$ represents a “null model” g trained on null string inputs $\emptyset$.

PVI requires a second “null model” of the same parameterization to be trained, but with the input X variable converted to an empty string to remove all information from the input variable that might aid the prediction of the output Y . Instances with lower PVI have been empirically validated as harder for human annotators to classify (Swayamdipta et al., 2020; Ethayarajh et al., 2022).

Loss $\uparrow$ The instance hardness literature has also explicitly considered loss as a proxy for instance-level complexity. Han et al. (2018) “selects its small-loss instances as the useful knowledge” during model training, where “useful” refers to generalizable patterns across instances in the dataset.

$$\text{Loss}(y_i | x_i) = y_i \log p(y_i) + (1 - y_i) \log (1 - p(y_i))$$

Thus, the loss literature suggests deep neural networks will learn easier and correct labels in earlier epochs before they become able to learn noisy or incorrect labels, and often employs dynamic subsampling or re-weighting (Arazo et al., 2019; Shen and Sanghavi, 2019). Items with higher prior loss are more likely to be misclassified in later epochs – as they were misclassified earlier.

3.4 Hardness meta-features

Boundary Distance (BD) $\downarrow$ BD was originally considered in the SVM literature (Tong and Koller,

Category	Metric	Computational Cost
Dynamic	PVI	$O(n \times E) + O(1)$
	Loss	$O(1)$
Static	PH	$O(n \times \mathcal{L}) + O(1)$
	TF	$O(1)$
IRT	IRT	$O(n \times E _{\text{IRT}}) + O(1)$
Hardness	BD	$O(1)$
Meta-features	SL	$O(1)$

Table 1: Complexity metrics and computational costs.

2001; Brinker, 2003). It can be seen as “the difficulty in separating the data points into their expected classes.” This complexity increases as the distance between a given point and the classification boundary shrinks (Lorena et al., 2024).

$$\text{BD}(y_{i,c}) = |y_c^* - y_{i,c}|$$

Here y_c^* refers to the class boundary between class $c \in C$ and its nearest neighboring class c^* .

We note that BD can be calculated *a priori* in the dataset and consequently does not vary across models or epochs for a dependent variable. Consideration of boundary points with high BD has led to the development of sampling techniques to increase the presence of minority classes (Walmsley et al., 2018; Chatzimpampas et al., 2023; Xie et al., 2023) and ensemble learning methods which can adaptively select the classifier that best fits the hardness of the instance (Dantas et al., 2019; Souza et al., 2019).

Sentence Length (SL) $\uparrow$ Due to the structured nature of language data, NLP studies typically leverage *a priori* linguistic features as a proxy for complexity (Soviany et al., 2022; Zhang et al., 2018). The most efficient and popular linguistic heuristic is sentence length, which is a simple count of the number of tokens in the input sequence:

$$\text{SL}(x_i) = ||x_i||$$

As sentence length grows (1) the number of possible grammatical parsing trees grows exponentially and (2) the chance of simple classifiers correctly guessing linguistic heuristics plummets (Spitkovsky et al., 2009).

4 Experiment

Having defined our metrics, we empirically examined their relationships through a text classification task. We trained 220 models on 2 different subsampled train sets across 5 dependent variables, computing and storing all aforementioned complexity

metrics. We then analyzed metric correlation as well as model performance and fairness on a held out set to determine patterns and differences.

A roadmap of the experimental procedure can be found in Figure 2. We consider “metric availability” to be a binary indication of whether a metric can be computed before or after model training begins. We note that equations for added computational cost indicate the additional runtime needed to generate each metric on top of the standard training pipeline.

4.1 Data

We consider five tasks across two datasets. For the first four tasks, we use the FairPsych NLP dataset of human-generated text responses with corresponding latent variable scores concerning a participant’s Anxiety, Numeracy, Subjective Literacy, and Trust in Physicians (Abbasi et al., 2021). The target variable is a real-valued score – i.e., average of multi-item responses, scaled between 0-1 – binarized for classification via median split.

Our fifth task is a separate depression detection task of transcribed speech from a clinical interview dataset with binary labels of Depression vs. Control provided by a clinical professional (Cotes et al., 2022). All data statistics and splits for all 5 tasks can be found in Appendix C and examples can be found in Appendix B.

4.2 Model Training and Storage

We chose three base neural network architectures as well as two transformer language models to train classifiers with varying hyperparameters. For each dependent variable, we trained 6 Feedforward Neural Networks (FFNs), 6 Convolutional Neural Networks (CNNs), and 6 Long-Short Term Memory networks (LSTMs). We fixed the learning rate at 1e-3 and vary the number of layers, nodes, and filters (when applicable). We also include 2 BERT and 2 RoBERTa models with learning rates of 3e-5 and 3e-6 (Devlin et al., 2019; Liu et al., 2019). Thus, we trained 22 models on 2 possible train splits (i.e., “hard” and “random”) across 5 dependent variables such that our entire model population included 220 models. We trained models for 15 epochs with an Adam optimizer using stochastic gradient descent and stored instance outputs and losses every epoch, which are sufficient statistics for most metrics.²

²Further details on computational considerations and training can be found in Appendices D and E, respectively.

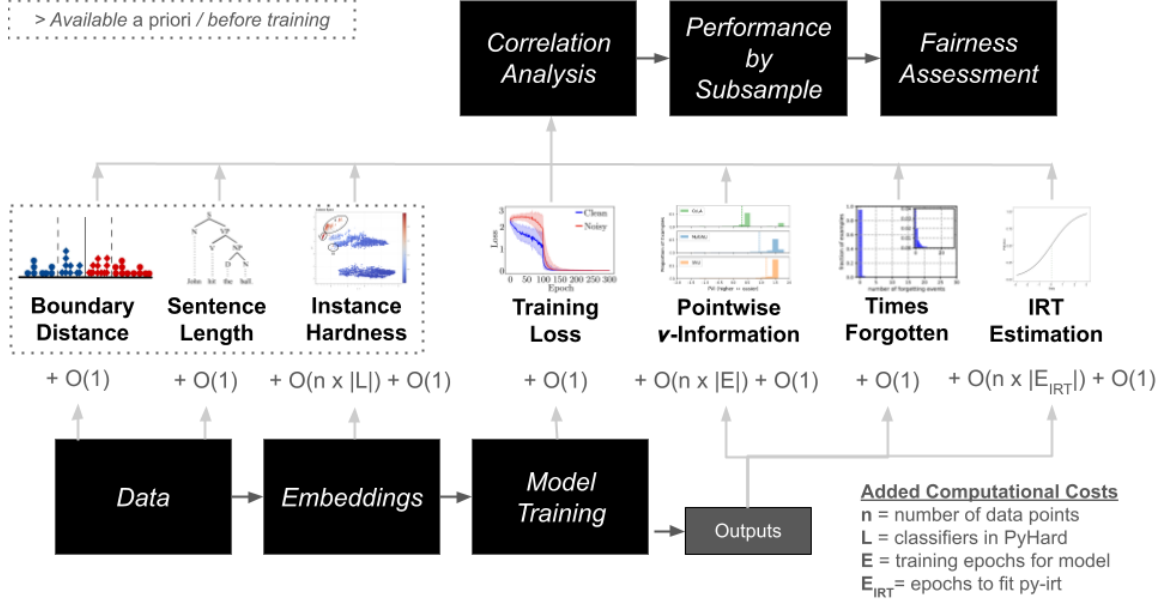


Figure 2: Experimental process diagram including the availability and added computational cost of each complexity metric calculation. Note that metrics available *a priori* / before model training are contained in the dotted box.

4.3 Experimental Procedure

4.3.1 Subsampling

We took two different training splits of the same dataset for each dependent variable. The “hard” train set was a stratified sample of 50% of instances with the highest BD. For the FairPsych data, we calculate BD as the difference between the true continuous variable value and the median split. For the depression detection data, we calculate BD as the Word Error Rate (WER) score between the automated transcription and the human gold standard as an alternative form of BD – i.e., $BD(i) = |0 - WER_i|$ since the human transcription error of every instance i is zero. The “random” train set was a random stratified sample of a similar length from the entire distribution. We used BD to identify hard examples because it is known *a priori* – as opposed to e.g., Loss, which is only known after training. There was approximately 45-50% overlap between the hard and random sample datasets, indicating a biased sample since the expected overlap from random draws would be 25% (i.e., $0.5^2 = 0.25$). We train our models (§4.2) with both splits and evaluate on our held-out test set.

4.3.2 Fairness

We also consider fairness throughout the machine learning pipeline. Lorena et al. (2024) recommends measuring the difference between protected and privileged demographic classes in their

respective distributions to ensure that instance-level complexity is equally measured *upstream* (i.e., during the representation phase before training) via **Kullback-Leibler Divergence (KLD)**:

$$D_{KL}(p||q) = \sum_{x \in X} p(x) \log \left(\frac{p(x)}{q(x)} \right)$$

where p and q are the respective distributions of the protected and privileged classes.

Downstream fairness refers to prediction and performance discrepancies of the trained models and can be assessed via **Disparate Impact (DI)**:

$$DI = \frac{p(\hat{y}=1|b)}{p(\hat{y}=1|a)}$$

for the probability p of predictions $\hat{y}$ of a feature with privileged class a and protected class b .

For both KLD and DI, larger values give evidence of fairness violations as they indicate more difference between privileged and protected classes (Lalor et al., 2024).

5 Results and Discussion

5.1 Complexity Difference by Train Set

Sampling on a single *a priori* available meta-feature creates datasets that are complex across several metrics. First, we examine how effective our BD split was in creating a difference in means for each complexity metric. We show this difference in complexity via a mean (M) difference analysis of each complexity metric across dependent variables and sampling strategies in Table 2.

We don't analyze the Anxiety task, as our subsampling did not create an $\alpha = 0.05$ significant difference between the "hard" ($M = 0.2544$) and the "random" ($M = 0.2574$) sets. Increases in BD were significant for Numeracy (-0.1034 , $p < 0.01$), Subjective Literacy (-0.0510 , $p < 0.01$), Trust in Physicians (-0.1006 , $p < 0.01$), and Word Error Rate (-0.0331 , $p < 0.01$). We also find significant differences in means for Loss, TF, PH, and IRT Difficulty across the four FairPsych tasks, and for TF and SL in the depression task. As we did not sample on these other complexity metrics, these results indicate some shared information that allows BD to create train sets of higher aggregate complexity across several metrics in different branches of the taxonomy.

	FairPsych				Interview
	Anx.	Lit.	Num.	Trust	Depr.
IH	0.000	0.000	0.034*	0.032*	0.083*
IRT	0.043	0.358	1.200*	0.530	1.577*
Loss	0.001	0.006	0.063*	0.027*	0.093*
SL	0.996	0.795	0.867	0.907	-2.187
TF	0.159*	0.409*	0.599*	0.361*	0.688*
BD	-0.004	-0.033*	-0.051*	-0.103*	-0.101*
PVI	0.022	-0.007	0.087	0.104	0.153

Table 2: Mean difference of metrics from Hard vs. Random subsampling on BD. *: $p < 0.05$, one-sided t-test with Bonferroni adjustment for multiple tests.

5.2 Correlation Analysis

Loss shares some inherent complexity feature(s) with several other metrics. In Figure 3, we show the aggregate correlation of our complexity metrics across all models and target variables, ordered by rank from highest to lowest. The x-axis shows the micro-averaged Spearman Correlation³ between each pair of metrics on the y-axis.

We note that Loss is present in 3 of the top 4 correlations, displaying some degree of related ranking to all Model-based metrics in the taxonomy except for PVI. Loss correlates moderately with TF ($\rho = 0.4236$) and IRT ($\rho = 0.4289$) as well as weakly with PH ($\rho = 0.3634$), which in turn correlate weakly with each other ($\rho = 0.3331$). This weak correlation between Loss and PH indicates the expected misclassification probability from PH is overlapping with the simpler calculation of the average Loss from the same model across epochs. Similarly, there exists a weak correlation

³The computation process for these correlations can be found in Appendix D.2

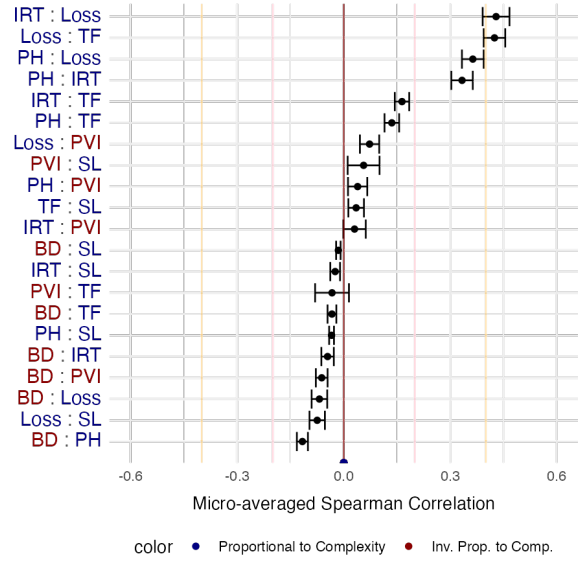


Figure 3: 95% CIs of Spearman Correlation averaged across all 5 dependent variables, train sets, and models. $\rho = 0$ marked by a dark red line. Thresholds for weak ($\rho = \pm 0.2$) and moderate ($\rho = \pm 0.4$) correlations are marked with light pink and orange lines, respectively. This plot is shown as a pairplot in Appendix F and broken down by task in Appendix G.

between the average Loss and the Difficulty values estimated by a 1PL IRT model. This additional IRT optimization problem is providing some of the same information that we are already computing in the model training process. Interestingly, PH and IRT are weakly correlated as well – although these optimizations are done in very different methods (i.e., empirical risk minimization vs. stochastic variational inference).

This analysis suggests that we can capture a substantial portion of Model-based complexity by simply storing the training Loss as a high level proxy for its shared complexity with PH, TF, and IRT Difficulty. Simply storing loss values is also more computationally-efficient and parsimonious than more theoretical techniques such as PyHard or IRT.

Our Hardness meta-features (i.e., BD and SL) are weakly correlated with other metrics in Figure 3, but BD is useful in sampling subsets that are significantly different along several metrics in Table 2. For example, while BD and Loss might not rank instances in exactly the same order, their relationship can still be used to identify similarly complex instances on aggregate. Future research should consider whether *a priori* metrics can similarly span the whole taxonomy as proxies for other metrics in aggregate filtering or instance-level re-weighting.

	AUC	Ability	Acc.	F1 Score
Anxiety	0.001	0.027	-0.005	0.055
Literacy	-0.056*	-0.856	-0.034	-0.051
Numeracy	-0.034*	-0.322	-0.008	0.052
Trust	-0.050*	-1.226*	-0.038	-0.071
Depr.	0.019*	-0.044	-0.013	-0.112*

Table 3: Mean difference in performance for Random vs. Hard BD subsampling strategies. Starred values are significant at $\alpha = 0.05$, two-sided t-test.

5.3 Performance Difference by Train Set

Subsampling on BD examples lowers model AUC. We want to examine whether models that saw more complex examples during the training process might perform better during inference. We measure test performance via several widely-used ML metrics (i.e., Accuracy, AUC, and F1 Score) as well as an Ability estimate provided by our IPL IRT model (Lalor et al., 2018). We display mean difference of performance metrics between each sampling strategy in Table 3 and overall performance in Appendix H.

We see a significant difference in performance for AUC, with almost all other performance metrics not showing significant difference across subsampling strategies. This AUC difference favors the randomly sampled data in all but the depression task. In short, training on more complex examples is typically harder for downstream prediction.

Measuring upstream fairness is sensitive to group imbalance. We note that model fairness is an issue for certain demographics and dependent variables, as shown both by our analysis of disparate impact (DI) as well as the proposed metric in Lorena et al. (2024) – which computes the Kullback-Leibler Divergence (KLD) between the protected group’s distribution and that of the privileged class. We find very similar distributions of complexity metrics for Sex, Race, and Income across dependent variables and complexity metrics, with KLD scores near zero across complexity metrics and dependent variables in Table 4. However, we notice a difference in complexity distributions for Seniors age 55+ and on education level for individuals who did not finish high school. We note that some of this divergence can be explained by the imbalanced data (Feng et al., 2018; Furundzic et al., 2017). The non-negative nature of KLD – i.e., as an asymmetric distance measure – ensures that it can only grow larger with distributional dif-

		Age	Sex	Race	Educ.	Inc.	ESL
Anxiety	BD	1.58	0.01	0.01	0.14	0.01	0.07
	IH	2.11	0.02	0.01	0.37	0.03	0.22
	IRT	2.22	0.02	0.02	0.25	0.02	0.15
	Loss	1.89	0.01	0.02	0.31	0.03	0.23
	PVI	2.19	0.02	0.02	0.22	0.04	0.15
	TF	1.85	0.03	0.02	0.33	0.02	0.15
	SL	1.65	0.03	0.05	0.24	0.02	0.15
Numeracy	BD	1.47	0.01	0.08	0.19	0.02	0.06
	IH	1.73	0.03	0.05	0.52	0.02	0.15
	IRT	1.75	0.04	0.14	0.59	0.02	0.11
	Loss	1.93	0.04	0.13	0.62	0.03	0.12
	PVI	2.37	0.04	0.14	0.74	0.03	0.15
	TF	2.52	0.03	0.08	0.55	0.02	0.13
	SL	2.21	0.02	0.06	0.30	0.02	0.18
Literacy	BD	1.93	0.02	0.02	0.42	0.01	0.12
	IH	1.80	0.03	0.01	0.32	0.02	0.14
	IRT	1.75	0.02	0.02	0.25	0.02	0.17
	Loss	1.38	0.02	0.02	0.31	0.02	0.15
	PVI	1.24	0.03	0.02	0.29	0.02	0.07
	TF	1.63	0.02	0.02	0.31	0.02	0.12
	SL	0.96	0.03	0.08	0.32	0.02	0.13
Depr.	BD	2.08	2.24	2.78	3.98	-	-
	IH	0.38	0.38	0.13	0.55	-	-
	IRT	0.29	0.32	0.07	0.42	-	-
	Loss	0.13	0.10	0.09	0.21	-	-
	PVI	0.27	0.30	0.11	0.31	-	-
	TF	0.08	0.09	0.07	0.21	-	-
	SL	0.08	0.08	0.05	0.09	-	-

Table 4: KLD between distributions of complexity values for the protected and privileged classes (larger values indicate greater difference). Explanation of missing values in the table can be found in Appendix I.2.

ferences, and this difference will result from a lack of coverage of the smaller distribution (Pan et al., 2005; Hochbaum and Pathria, 1998). The lack of coverage is likely related to the size of the protected class for Age (1.93%) and Education (1.32%) being very small in the FairPsych dataset. The depression data displays some imbalance for Age (8.26%) and Education (26.31%), but we also see large KLD values for the more balanced Race (36.00%) and Sex (50.37%) demographics. Thus, our WER measurement of BD is biased, which is consistent with previous research on unequal performance of automated transcription methods (Koenecke et al., 2020). Future research should be mindful of how the choice of fairness metric impacts complexity comparisons across demographic groups.

Subsampling on complexity does not impact downstream fairness. In Figure 4, we examine whether our sampling strategy impacts model fairness as measured by DI confidence intervals – We also note that the fairness literature considers DI scores below 0.8 or above 1.2 – marked with red

lines – to disproportionately affect predictions for the protected class, so we bold any CIs that lie fully outside these suggested thresholds (Lalor et al., 2022). We note that none of our 95% CIs lie completely above 1.2 such that models do not predict the protected class significantly more across demographics and dependent variables. Although several CIs lie fully below 0.8 (e.g., Numeracy-Race, Subjective Literacy-Income), there is only one significant difference between CIs of different sampling strategies (i.e., Depression-Age) so fairness concerns also exist for models trained on random data. Notably, in 17 out of the 27 cases presented in Figure 4, the hard CIs are either smaller than the random CIs, or comparable-sized but shifted towards the center (i.e., indicating less disparate impact). This is important from a risk management perspective since both the extent of downstream bias and its variance are key considerations. Thus, we should not expect sampling on complexity to systematically bias against protected groups.

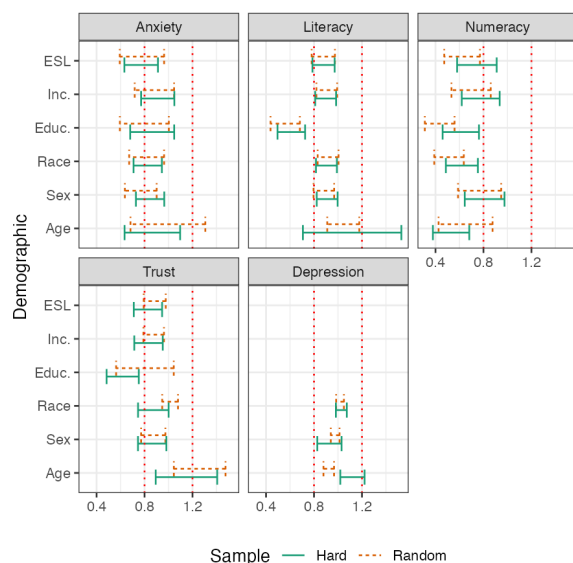


Figure 4: 95% Confidence Intervals for Disparate Impact across demographics. All CIs overlap indicating no significant difference at $\alpha = 0.05$. Red values indicate DI intervals fully below the “0.8 rule.” See Appendix I.2 for information on missing values.

6 Conclusion

We computed various complexity metrics for text classification and analyzed their correlation across dependent variables. These findings supported updates to the current taxonomy on classification complexity to consider metric availability and computational efficiency. We found that simply storing

training Loss captured similar complexity of computationally intensive model-based metrics such as IRT Difficulty and Instance Hardness. Further, sampling on a single *a priori* available meta-feature created datasets that were complex across separate metrics. We only found one difference in upstream fairness and no downstream impact from employing these complexity metrics in a sampling strategy.

7 Limitations

There are limitations that can inform future work on instance-level complexity in NLP tasks. We note that the updated taxonomy is by no means exhaustive and only applies to classification tasks. We limit our empirical examination to the healthcare setting, but future studies might compare complexity relationships measured on text from other applied areas. While 4 of our tasks provide a ground-truth latent continuous variable as a built-in Hardness meta-feature (via BD), future research should examine other BD measures from types of input noise like WER, provided that they can be fairly applied across demographics.

Recent work has also started to explore using LLMs to facilitate instance hardness estimation. Future work can further investigate the advantages of LLMs in terms of potential efficiency and robustness gains (Lu et al., 2023; Saha et al., 2022). While linguistics research may offer new Hardness meta-features from further understanding of grammar and syntax, we might also expect new complexity metrics to arise in the Model-based section of the taxonomy from new architectures and techniques such as in-context learning (Lu et al., 2023; Valmeekam et al., 2022), retrieval augmented generation (Jeong et al., 2024; Chen et al., 2024), or graph learning (Wang et al., 2024). Our updates to the taxonomy are not meant to be exhaustive, and we encourage future research to continue exploring the many facets of complexity.

Acknowledgments

This work was partially supported by the National Science Foundation under Grants IIS-2403438, IIS-2240347, and IIS-2039915.

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A Metric Literature Search

Table 5 lists the papers included as a result of our literature search. Each representative metric used in our experiment was chosen from a group of several similar metrics provided in the “Related Metric” column of the table.

B Data Example Instances

We provide example instances from each task in Table 6. “Low” and “High” observations were randomly sampled from the lower and upper quartiles,

Complexity Class	Representative Metric	Related Metric
Hardness meta-features	Boundary Distance (Ho and Basu, 2002)	SVM Margin (Malisiewicz et al., 2011; Tong and Koller, 2001; Brinker, 2003) Influence Functions (Koh and Liang, 2017) k-Center (Sener and Savarese, 2018) Outlier Detection Algorithms (Hodge and Austin, 2004) Utterance Length (Amodei et al., 2016)
	Sentence Length (Soviany et al., 2022; Zhang et al., 2018)	Word Rarity (Zhang et al., 2018) Number of Coordinating Conjunctions (Kocmi and Bojar, 2017) Sentence Nesting (Zaremba and Sutskever, 2014)
Item Response Theory	IRT Difficulty (Martínez-Plumed et al., 2019; Lalor et al., 2018)	N / A
Static Hardness	Instance Hardness (Smith et al., 2014)	Representation Bias (Le Bras et al., 2020) Eigenvector Density (Gong et al., 2021) Vector Norm (Liu et al., 2020) Neighborhood Instance Selection (Olvera-López et al., 2010)
	Times Forgotten (Toneva et al., 2019)	N / A
Dynamic Hardness	Loss (Han et al., 2018; Arazo et al., 2019; Shen and Sanghavi, 2019)	Confidence (Zhang et al., 2018; Swayamdipta et al., 2020; Saha et al., 2022) Area Under Cost Curve (Prudêncio and Castor, 2014) Uncertainty (Joshi et al., 2009) Variability (Chang et al., 2017; Swayamdipta et al., 2020) Informativeness (Jafarpour et al., 2021) Gradient (Ren et al., 2018) Gradient-based Importance (Vodrahalli et al., 2018) Informativeness (Jafarpour et al., 2021) Information Theoretic (Zhang et al., 2020)
	Pointwise v-Information (Ethayarajh et al., 2022)	Shapley Values (Ghorbani and Zou, 2019) In-context PVI (Lu et al., 2023)

Table 5: Metrics considered during the literature review along with their inclusion or similar metric

respectively. We note that scores for the depression task are calculated from the Word Error Rate (WER) between the Amazon Web Services (AWS) automated translation and a human-transcribed gold standard. Unlike the four FairPsych tasks, there is no common prompt for the depression task since data comes from unstructured conversation.

C Data Splits

For both datasets, we assign individuals randomly to one of the train-random, train-hard, validation, or testing splits on a 70-20-10% split. This split is done before subsampling on BD such that the “Rand.” and “Train” consist of half of this full train set (i.e., 35% of the total dataset). While there is overlap in individuals between train-random and train-hard, none of these them appear in validation or test data. Counts of individuals for each split can be found in Table 7.

D Computational Considerations

D.1 Complexity Metrics

Since BD and SL are available *a priori*, we made simple calculations from the input data. Loss was generated via the model training process and required no further calculations. For TF, we counted any time an incorrect instance was correct in the previous epoch (Toneva et al., 2019). To calculate PVI, we followed guidance from prior work (Ethayarajh et al., 2022). Specifically, we trained a “null model” identical to the primary model, but with no text input (i.e., an empty string), incurring an additional $O(n \times |E|)$ runtime for the second model. PVI was then calculated from the difference in entropy between the output probabilities of the primary model and those of the null model.

To calculate PH, we used the PyHard package (Paiva et al., 2021) to compute a single instance hardness statistic for each observation. PyHard computes scores across seven diverse classifiers: “Bagging, Gradient Boosting, Support Vector Machine (both linear and RBF kernels), Logistic Regression, Multilayer Perceptron, and Random Forest” and leverages instance space analysis (ISA), an embedding technique that combines information from meta-features and candidate algorithms to generate the final aggregate score (Paiva et al., 2021). Since PyHard only takes numerical data as input, we encoded text documents as BERT embeddings (Devlin et al., 2019). There was still a relatively high additional computational cost $O(n)$ to

generate BERT embeddings and then a $O(n \times |\mathcal{L}|)$ runtime to train the simple classifiers, where PyHard’s default configuration uses an ensemble of $c = 7$ classifiers on 5 fold cross-validation. The wall-clock time on each dataset of $N \approx 4,200$ observations was typically 45 minutes to an hour for each dependent variable.

To learn IRT difficulty with our dataset size ($N \approx 1,700$), we leveraged the py-irt package (Lalor and Rodriguez, 2023). We created a dichotomous response matrix for each of our 176 models and ran 8 IRT estimations for model ability and item difficulty across the 2 train sets and 4 dependent variables. Although the runtime complexity for running the stochastic variational inference estimation adds an extra $O(n \times |E_{IRT}|)$, py-irt’s GPU-accelerated training was accomplished in a wall-clock time of less than a minute. We can visualize model performance by model class and subsampling strategy for each dependent variable in Table 8 in the appendices. This shows us variation in performance across model types, ensuring variety in predictions for our model-based instance hardness classifications.

D.2 Spearman Correlations

We compared the instance-level correlations across models and dependent variables. We emphasize that our various complexity metrics are necessarily calculated at different levels of hierarchy in the experiment (Figure 1). Thus, calculations of Loss and PVI – which are computed each epoch – are averaged for a given model. While PH, TF, and IRT Difficulty are all calculated at the model-level, we also note that BD is known *a priori* from the data such that this metric exists at the dependent variable-level and displays no variance across models. All correlations are calculated at the instance-level and results for models and dependent variables are computed from micro-averaging instance-level correlations. Lastly, we are considering Spearman Rank Correlations since we are more concerned with the rank / ordering of the instance IDs, especially since the magnitude of some complexity metrics have no unitary interpretation (e.g., BD, IRT Difficulty).

E Model Parameters and Training

For each dependent variable, we trained models by grid searching the parameters in Table 9. We also give the total possible size of each model as “Max. Param.” and the average time to train all models for

Task	Prompt	Quartile	Text	Score
<i>Anxiety</i>	In a few sentences, please describe what makes you most anxious or worried visiting the doctor's office.	High	It depends on the doctor and the reason for the visit. If I have to go to the gynecologist, it's stressful due to the extreme invasion of privacy. I'm normally not anxious when going to my primary care doctor, unless I have to provide a blood sample. I have an extreme fear of needles.	0.9762
		Low	Right now I have nothing to fear. I did get a diagnosis of high blood pressure but that was due to my job and then eventual termination of said job. I'm also unhappy in my relationship so that may caused a spike.	0.1429
<i>Numeracy</i>	In a few sentences, please describe an experience in your life that demonstrated your knowledge of health or medical issues.	High	I have ran many marathons successfully and taken care of all recovery that had to do with it without any outside help.	1.0000
		Low	I was recently told that I needed a root canal. I was told why I needed one and the steps that are taken to complete the procedure.	0.3571
<i>Subjective Literacy</i>	Regarding all the questions you just answered, to what degree do you feel you have capacity to obtain, process, and understand basic health information and services needed to make appropriate health decisions? Please explain you answer in a few sentences.	High	As a retired RN, I am well versed in the field of healthcare. I am able to research any topics or findings I am unfamiliar with and, I hold health care providers accountable for the quality of care they provide.	0.9167
		Low	a little because some terms i dont know. all i know is spina bifida and paralysis and water on the brain is what i have.	0.4278
<i>Trust in Physicians</i>	In a few sentences, please explain the reasons why you trust or distrust your primary care physician. If you do not have a primary care physician, please answer in regard to doctors in general.	High	My doctor has a great reputation with his patients. He's been a great doctor and has treatment effectively when other doctors have not been successful. He makes sure I understand everything before leaving his office.	0.9600
		Low	In all honesty, I do not trust my doctor at all. It seems as if most doctors in general mostly just care about their own convenience and making money. Even though the job itself is to help people, to them it's still just a job to them at the end of the day.	0.36
<i>Depression</i>	<i>None / in-conversation</i>	High	quite a few bigger fears. Um	0.2506
		Low	Um I am a school psychologist so work in the schools, My father um	0.1450

Table 6: Example text responses to question prompts from high and low scored individuals on multi-item scales used for each dependent variable in the FairPsych and depression interview datasets (Cotes et al., 2022; Abbasi et al., 2021).

Dep. Var.	Train		Val.	Test
	Rand.	Hard		
Anxiety	2,938	2,904	841	1,678
Numeracy	2,969	2,778	848	1,698
Literacy	2,976	2,960	850	1,701
Trust	2,974	2,464	850	1,699
Depression	667	670	413	434

Table 7: Number of observations in each split across dependent variables

		AUC	Acc.	F1 Score	IRT
<i>Anxiety</i>	FFN	60.27	56.37	66.91	1.58
	CNN	63.02	59.05	60.52	1.93
	LSTM	65.70	61.08	67.21	2.29
	BERT	70.71	65.37	67.11	2.86
	RoBERTa	71.18	65.85	70.30	2.83
<i>Numeracy</i>	FFN	67.84	63.89	63.82	2.90
	CNN	71.00	62.77	54.11	3.32
	LSTM	69.96	64.60	64.48	3.58
	BERT	75.92	68.31	68.35	4.19
	RoBERTa	76.08	68.37	69.54	4.10
<i>Literacy</i>	FFN	74.10	67.48	71.61	2.96
	CNN	74.97	68.66	70.85	3.23
	LSTM	75.99	69.90	71.77	3.48
	BERT	79.16	72.01	73.07	3.70
	RoBERTa	79.48	70.31	74.36	3.81
<i>Trust</i>	FFN	78.73	72.10	70.31	4.00
	CNN	76.49	70.45	66.06	3.67
	LSTM	78.75	72.27	69.63	4.20
	BERT	84.04	76.63	73.55	5.39
	RoBERTa	84.69	76.33	75.27	5.45
<i>Depression</i>	FFN	52.37	56.00	71.42	1.16
	CNN	54.48	55.80	71.46	1.06
	LSTM	56.06	56.00	70.58	1.30
	BERT	55.15	55.60	71.46	0.84
	RoBERTa	59.30	56.59	62.52	1.12

Table 8: Performance metrics by dependent variable and model type. Performance levels are comparable for FairPsych and clinical Depression benchmarks, even while using a 50% subsample (Abbasi et al., 2021; Qin et al., 2024).

a given dependent variable as “Total Wall Clock.”

All training was done on a University HTCondor system and models trained on GPU are marked with an asterisk. Beyond the 4 GPU hours above, we note another 4 hours for training “null models” for PVI, another 1 hour for generating BERT embeddings for PyHard, and another 1 minute for running py-irt for a total of ~ 9 GPU hours.

F Correlation Pairplot

In Figure 5, we include a pairplot of correlations micro-averaged across tasks and data samples for comparison with the confidence interval plot shown in Figure 3.

G Complexity Metric Correlations by Task

In Figure 6, we replicate the micro-averaged Spearman Correlations shown in Figure 3 but calculated on subsamples of data from each dependent variable / task. We order the metrics on the y-axis in the same way as the in Figure 3’s aggregate results to highlight variation within and across tasks but similar overall trends, especially in the top four large positive correlations seen in the main results (i.e., IRT : Loss, Loss : TF, PH : Loss, and PH : IRT).

H Performance Metric Distribution

Figure 7 plots the distributions of each performance metric achieved by models across dependent variables. IRT Ability is min-max scaled to also fall between 0 and 1 by subtracting the minimum value and dividing by the range. This is done for the sake of comparison with the other metrics on the same range, and that the range for raw Ability scores is (-2.67, 5.45). Since IRT Ability scores typically fall into a (-6, 6) range, we do not seem to be masking any extreme outliers.

I Demographic Thresholds and Missing Values

I.1 Population Thresholds

Aside from a few individuals who put null text strings for one of the tasks, we should consider the population of individuals to be almost entirely the same across tasks for the FairPsych data. Thus, the same individual might be in a different splits or hardness subsample across tasks Even if dependent variables were not assumed to be independent, we

Model	# Nodes	# Layers	# Filters	Kernel Size	Max. Param.	Learn. Rate	Total Wall Clock
FFN	64, 128	1, 3, 5			90,786	1e-3	~25 min.
CNN		1, 3, 5	16, 32	5	28,002	1e-3	~15 min.
LSTM	32, 64	1, 3, 5			255,682	1e-3	~120 min.
BERT					110M	3e-5, 3e-6	~25 min. *
RoBERTa					123M	3e-5, 3e-6	~25 min. *

Table 9: Parameters and wall clock times for grid search used to the train models in this project. Stars indicate models that had accelerated training on GPU.

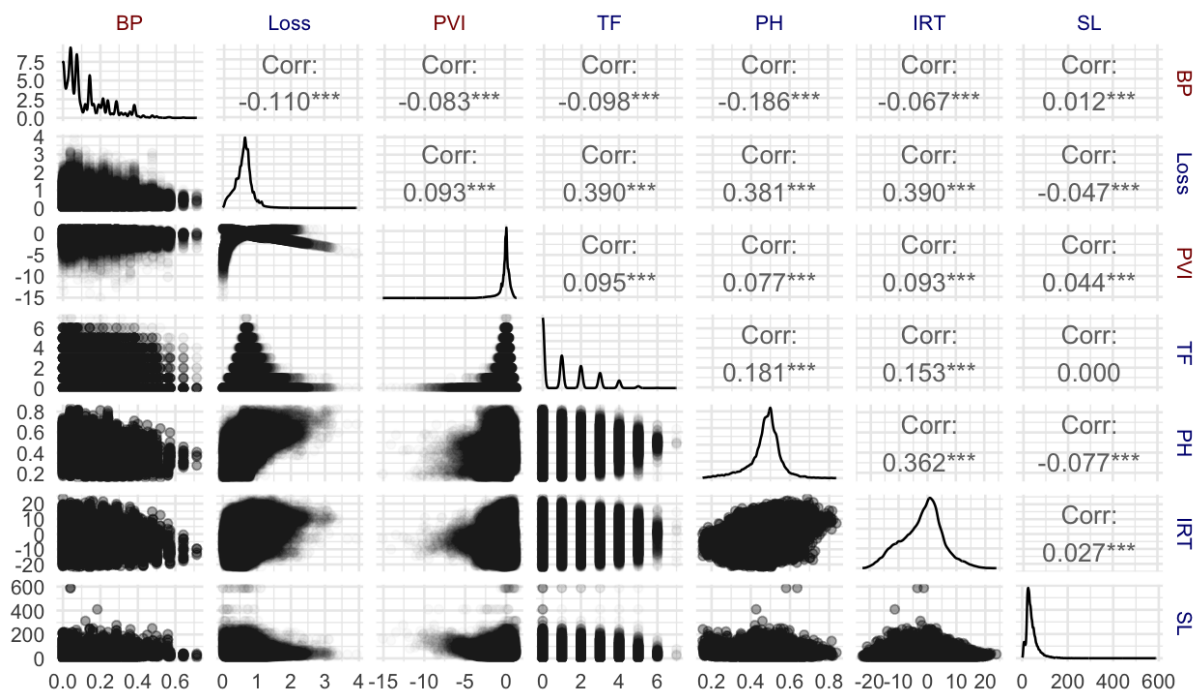


Figure 5: Pairplot of micro-correlations calculated across all dependent variables and samples. While lower plots show linear scatter plots, the upper correlation values are Spearman Correlations to match the analysis in Figure 3. Distributions are shown on the diagonals.

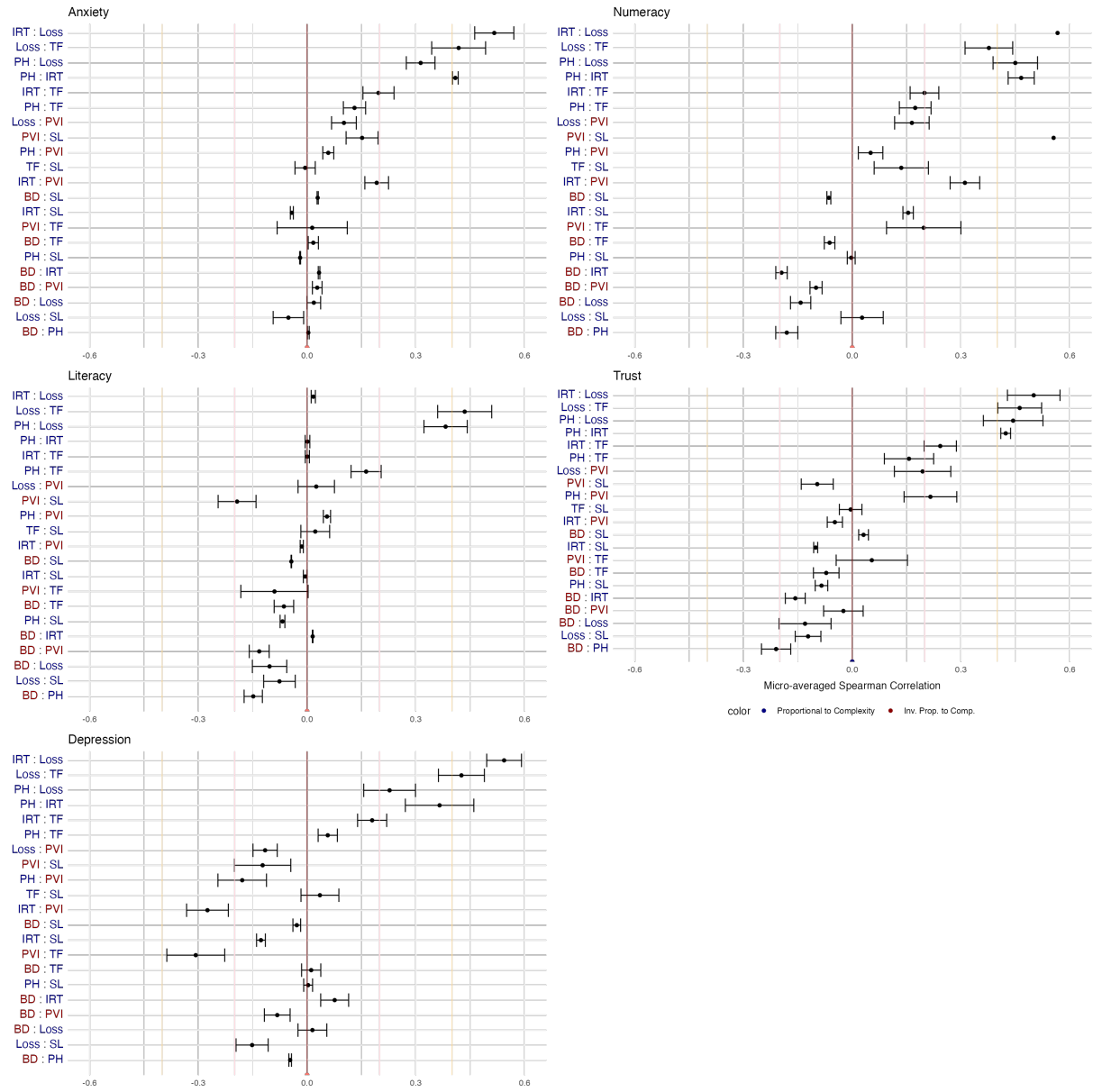


Figure 6: Similar to Figure 3, we provide 95% CIs of Spearman Correlation for each dependent variables micro-averaged across train sets and models. Again, we mark $\rho = 0$ with a dark red line and include thresholds for weak ($\rho = \pm 0.2$) and moderate ($\rho = \pm 0.4$) correlations as light pink and orange lines, respectively.

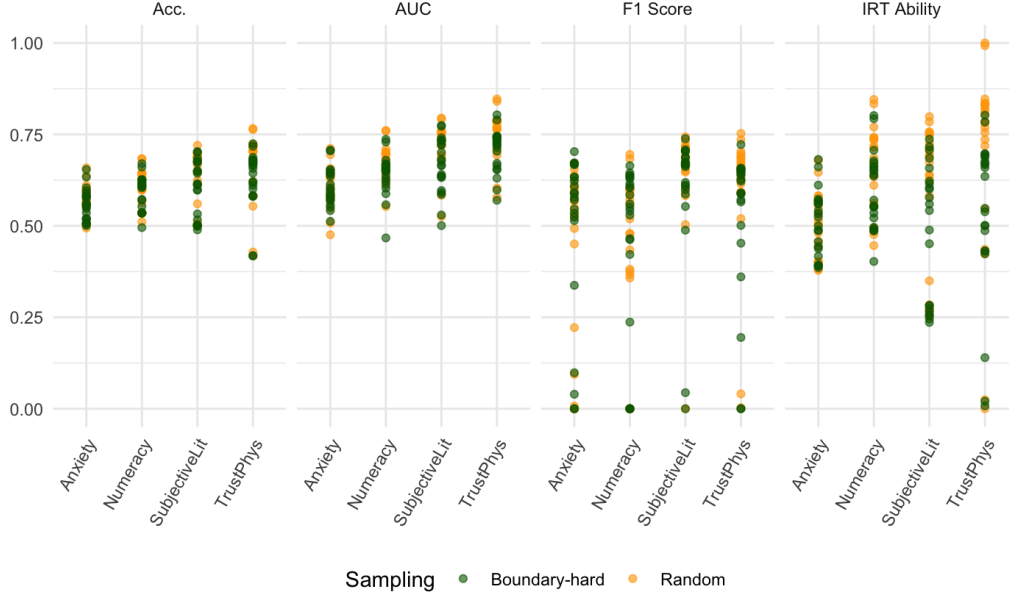


Figure 7: Distribution of scores for common performance metrics achieved by models across each task. Note that IRT Ability is min-max scaled between 0-1 for the sake of comparison.

use separate models for each task such that there is no leakage if e.g., an individual ends up in the train set for Numeracy and the test set for Anxiety. Lastly, it is perfectly reasonable that someone might have e.g., borderline low Literacy and very low Numeracy – both with respect to the median split. Thus, the former high BD value might place the individual in train-hard for Literacy and train-random for Numeracy. Moreover, some individuals might appear twice or not at all since the subsampling is done randomly. Again, the separating models by task prevents leakage, but increases demographic variation across data splits and tasks.

Since the clinical depression data only has one task and a one-to-many mapping of individual to document, we must split at the individual level. The expensive nature of clinical interviews leaves us with a dataset that is sparse with respect to both individuals and utterances. Thus, we have a smaller population of individuals with fewer sources of demographic variation across data splits. Additionally, this study uses different a different sampling procedure so certain demographic groups from the FairPsych dataset are systematically missing from the clinical depression data (e.g., English-as-Second Language).

Due to these differences in data sampling procedures, it is reasonable to consider all FairPsych tasks and splits to be from one population and the depression task to be from a different popula-

Demo.	FairPsych	Depression
Age	≥ 65	≥ 53
Sex	Non-male	Non-male
Race	Non-White	Non-White
Educ.	$\leq$ High school graduate	$\leq$ Some college or trade / vocational school
Inc.	$\leq$ \$54,999	NA
ESL	Yes	NA

Table 10: Threshold values to be considered part of the protected class in each dataset. Note that all cut-off values are inclusive with directions indicated when applicable.

tion. Examination of the distribution of Education Level and Age lend further evidence to the existence of two separate populations. Therefore, we consider different thresholds for what constitutes a “protected group” in each population. We borrow threshold values from Abbasi et al. (2021) for the FairPsych data and propose reasonable cut-offs for the clinical depression demographics. The criteria for being in a protected group for each dataset can be found in Table

I.2 Missing Values

Here, we explain the missing values in the Kullback-Leibler Divergence (KLD) and Disparate Impact (DI) calculations in Table 4 and Figure 4.

First, we note that the clinical depression dataset does not provide Income data, so we lose this col-

umn for KLD and DI. Further, it only considers native English speakers, so we have no English-as-Second-Language participants and again have NA values for both KL Divergence and DI. Second, our *upstream* fairness measurement of KLD is calculated from complexity measurements on the entire datasets / populations, but *downstream* results such as DI consider output predictions from the deployed models. Thus, our populations are reduced to the subsample of individuals randomly assigned to the test / held-out sets.

We employ a 70-20-10 data split for both datasets; see Appendix C above for an explanation of data sampling procedures. Since there is a one-to-one mapping between individuals and text documents in the FairPsych data, the 10% split for the test set still comprises 1,700 individuals for each task. However, the depression dataset has a one-to-many mapping of individuals to documents and only has 40 individuals total in the dataset. While we predict at the utterance / document-level, data splits are done at the individual-level to prevent leakage. Thus, the test set contains 434 documents but from only 6 different individuals and we are unlikely to see individuals from an imbalanced protected class appear in this data. Our small test set does not have members of the protected class for Education Level, so we cannot calculate downstream DI scores from the test set, despite having upstream calculations of KLD on the entire dataset.